Paul Lee

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Education

University of Virginia Charlottesville, VA
Bachelor of Arts in Computer Science, GPA: 3.92 *Aug. 2024 – May 2028*
Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Computer Systems & Organization, Software Development Essentials, Software Engineering, Compilers, LLMs and Software Engineering

Experience

Software Engineer Intern May 2026 – Aug. 2026
Zeta Associates (Lockheed Martin Subsidiary) *Fairfax, VA*

- Built an autonomous GitLab AI agent in Python that parses repository URLs, constructs structural codebase context, and answers complex, codebase-wide technical questions
- Engineered a Kubernetes diagnostic agent that autonomously inspects live cluster state, isolates failing pods and misconfigured deployments, and recommends remediation steps to reduce manual triage effort
- Prototyped an autonomous merge-request reviewer that sandboxes MRs, runs build/test suites, and iteratively patches failures over configurable retry loops; opened follow-up MRs with Markdown reports on fixes applied and issues needing review
- Benchmarked agentic AI frameworks (CrewAI, Mastra, Agno) for feasibility in an air-gapped, classified environment, informing the team's tooling adoption strategy

Technical Skills

Languages: Python, TypeScript, Java, JavaScript, SQL, C, HTML/CSS
Frameworks & Libraries: Mastra, CrewAI, Agno, AutoGen, n8n, React Native, Django, JavaFX
Cloud & Tools: Kubernetes, Docker, AWS S3, Heroku, Supabase, PostgreSQL, Git, VS Code
Practices: Agentic AI, MCP, DevOps, CI/CD Monitoring, Agile/SCRUM, MVC Architecture, RESTful APIs

Projects

UVA Sustainability Marketplace | *Django, Python, PostgreSQL, AWS S3, Heroku* Aug. 2025 – Dec. 2025

- Collaborated in an Agile team of 4 to design and build a full-stack sustainability platform for the UVA community using Django and PostgreSQL
- Built a real-time notification engine with triggers for item requests, approval workflows, and peer-to-peer messaging
- Implemented a transaction-state and buyer-request system letting sellers dynamically manage item availability and filter listings via a custom dashboard
- Designed admin security tools, including a session-level user suspension framework and a centralized moderator dashboard for role-based governance
- Deployed the application to Heroku with AWS S3 media storage and Google OAuth, supporting team-wide releases and code quality standards

Course Review System | *Java, JavaFX, SQLite, MVC, Git* April 2025 – May 2025

- Built a full-stack course evaluation platform in a team of 3, owning database design end-to-end and enabling students to create, read, update, and delete course reviews
- Designed a normalized relational schema and optimized SQL queries in SQLite, improving query performance through proper indexing
- Developed a maintainable JavaFX interface using FXML and controllers following MVC principles, and participated in code reviews to uphold code quality

Hoos In Love Mobile Application | *React Native, JavaScript, PostgreSQL, Supabase* Oct. 2025 – Present

- Building a mobile dating platform for 500+ UVA students with end-to-end ownership from design through planned deployment; architected a relational database with 14+ normalized tables supporting matching, messaging, and user preferences
- Researched and implemented a collaborative-filtering matching algorithm to rank potential matches by user preferences, interests, and compatibility scores
- Built authentication flows and profile interfaces from Figma designs in React Native, including multi-photo upload, bio editing, and preference management
- Designed a PostgreSQL schema for users, matches, swipes, messages, and chats with cascading-delete constraints and referential integrity